

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 3: Research Methodology

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3 Research Methodology

This chapter specifies the empirical design used to compare EndorseRank and Adaptive Weighted PageRank (AWP) as social reputation scores on Arbitrum One. The design covers notation, data sources, a three-phase collection strategy, pipeline architecture, event decoding, graph construction algorithms, validation proxies, statistical tests, computational benchmarks, scaling and robustness extensions, reproducibility, and ethics. The primary comparison is between EndorseRank on the ERC-20 allowance graph and AWP on the time-decayed transfer graph ^{[16],[49]}. GainFlow PageRank (GF-PR) appears only as a supplementary outcome-native ceiling reference and is not treated as a peer social method.

The methodological stance follows construct (domain) alignment: reputation scores are continuous rankings, and validation uses observable wallet-level proxies rather than binary default labels ^{[14],[16]}. In measurement terms, each ranking method is an indicator of an unobserved reputation construct; proxies are alternative indicators chosen for domain intuition (inbound economic activity, received endorsements, risk-avoiding trading behavior, activity stability, and trading success). Alignment statistics ask whether the ordering induced by a method co-varies with the ordering induced by each proxy on the same wallets. This approach is appropriate when ground-truth loan defaults are sparse and when the scientific claim is comparative—EndorseRank versus AWP—rather than absolute prediction of an external credit bureau label.

All procedures operate on publicly available blockchain logs within a fixed observation window, with deterministic seeds and an open evaluation pipeline so that reported tables can be regenerated from cached parquet. The chapter is organized to move from symbols and data acquisition, through graph and proxy formalisms, to evaluation statistics, computational protocol, and research ethics. Readers primarily interested in empirical outcomes may skim Sections 3.1–3.5 and focus on Sections 3.6–3.10; readers implementing a replication should follow the full pipeline description and

Section 3.13.

Four principles govern every operational choice.

- **Same-seed comparability:** EndorseRank and AWP are always scored on identical wallet sets so that differences cannot be attributed to sampling inconsistency.
- **Pre-registration by configuration:** observation window, topic hashes, damping, decay parameters, minimum closes, and benchmark seeds are fixed in configuration before interpreting results.
- **Separation of construction and timing:** graph construction is performed once per cohort; computational benchmarks time only the PageRank solve.
- **Primary versus supplementary scope:** claims C1–C4 and hypotheses H1–H3 concern EndorseRank and AWP; GF-PR enters only SRQ1 as an outcome-native ceiling.

3.1 Notation

Table 3.1 summarizes the operational symbols used throughout this chapter and in Chapters 4–5 (a compact front-matter summary appears in Table 2). Vectors are bold; sets are calligraphic when helpful. Wallet addresses are treated as opaque identifiers and are never linked to off-chain identities.

Unless otherwise stated, PageRank is the weighted formulation of Page et al.^[49]: at iteration t ,

$$\mathbf{r}^{(t+1)} = (1 - d) \boldsymbol{\pi} + d P^\top \mathbf{r}^{(t)} + d (\mathbf{r}^{(t)\top} \mathbf{1}_{\text{dang}}) \boldsymbol{\pi}, \quad (3.1)$$

where P is the row-stochastic transition matrix induced by positive edge weights, $\boldsymbol{\pi}$ is the uniform teleportation distribution over nodes, and $\mathbf{1}_{\text{dang}}$ marks dangling nodes (zero out-strength). Explicitly, if $w(u, v)$ denotes the weight of the directed edge $u \rightarrow v$ and $s(u) = \sum_v w(u, v)$ is the out-strength,

Table 3.1: Notation for graphs, scores, proxies, and parameters.

Symbol	Meaning
$\mathcal{W}$	Matched wallet cohort ($ \mathcal{W} = n = 5,521$)
$\mathcal{W}_{\text{pool}}$	Expanded wallet pool ($ \mathcal{W}_{\text{pool}} = N = 31,612$)
$G_{\text{approve}} = (V_A, E_A, w_A)$	EndorseRank allowance graph
$G_{\text{transfer}} = (V_T, E_T, w_T)$	AWP transfer graph
$G_{\text{GF-PR}}$	Supplementary GF-PR star graph
$w_A(u, v)$	Latest allowance weight, owner $u \rightarrow$ spender v
$w_T(u, v)$	Time-decayed transfer weight, sender $u \rightarrow$ receiver v
$\mathbf{s}_M$	Reputation score vector for method M
$\mathbf{p}_P$	Proxy value vector for proxy P
d	PageRank damping factor (default 0.85)
τ_{tol}	PageRank convergence tolerance (10^{-8})
I_{max}	Maximum PageRank iterations (300)
$\sigma(\Delta t)$	Logistic time-decay on transfer age
k, t_0	Decay steepness (0.01) and midpoint (180 days)
Δt	Age of an event in days before observation end
$\varepsilon_{\text{loss}}, \varepsilon_{\text{liq}}$	GF-PR sink additives (0.1, 1.0)
ρ	Spearman’s rank correlation
τ	Kendall’s rank correlation
T_{obs}	Observation end timestamp (2026-05-31 UTC)

then

$$P_{u,v} = \begin{cases} w(u,v)/s(u) & \text{if } s(u) > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

Dangling mass is redistributed uniformly through the teleportation term, which is the standard treatment in sparse web and transaction graphs. Iteration stops when $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \tau_{\text{tol}}$ or after I_{max} iterations. Scores are renormalized to sum to one over reachable nodes. Dense ranks for alignment are assigned by sorting scores descending (rank 1 = highest score), with wallet address as a deterministic tie-break when scores are equal.

The damping factor d interpolates between pure graph propagation ($d \rightarrow 1$) and uniform scores ($d \rightarrow 0$). The default $d = 0.85$ follows the classical web setting and the AWP baseline of Do et al.^[16]. Robustness sweeps vary d while holding edge weights fixed (Section 3.12). Tolerance $\tau_{\text{tol}} = 10^{-8}$ and $I_{\text{max}} = 300$ are intentionally strict relative to interactive analytics settings, so that runtime differences reflect graph structure rather than early stopping.

3.2 Data sources and sampling

This section fixes the empirical universe: which chain, which time window, which public tables, which event types, and which wallets enter the primary sample. Downstream graph construction and proxy formulas never invent events outside this universe.

- **Primary chain:** All on-chain events are drawn from Arbitrum One (chain ID 42161), an Ethereum Layer-2 rollup with dense DeFi activity and public log availability^[53]. Restricting to a single chain avoids cross-chain address aliasing—the same private key controlling addresses on multiple chains would otherwise appear as unrelated nodes—and keeps gas-cost and latency regimes comparable across wallets. Arbitrum One is also the deployment venue for GMX V2’s high-volume perpetual markets, which supply wallet-attributed realized PnL at a density that lending defaults do not match within scope.
- **Observation window:** The study window is 2025-12-01 00:00:00 UTC through 2026-05-

31 23:59:59 UTC (six calendar months). Exclusive end timestamps in SQL use 2026-06-01 00:00:00 UTC so that the inclusive calendar end is 2026-05-31. This window is long enough to accumulate multiple GMX closes per active trader—the minimum-close filter requires at least three—while remaining tractable for monthly-batched reputation extracts under BigQuery byte budgets. AWP time-decay is anchored at observation end $T_{\text{obs}} = 2026-05-31$ 23:59:59 UTC, so $\Delta t = 0$ for events at the window’s close and $\Delta t \approx 181$ days for events at the window’s open.

- **BigQuery source:** Event logs are read from the public dataset

`bigquery-public-data.goog_blockchain_arbitrum_one_us.logs.`

Each row corresponds to a contract log entry with block timestamp, emitting address, topics, and data. Queries filter on `block_timestamp`, contract address (when applicable), and event topic hashes. The public dataset removes the need for a private archive node while remaining auditable: any researcher with Google Cloud access can re-issue the same SQL against the same table. SQL templates, dry-run cost guardrails, and acquisition metadata are documented in Appendix 8.1.

- **Reputation events:** EndorseRank uses ERC-20 Approval logs ^[18], including allowances established via EIP-2612 permit flows that ultimately emit the same Approval event ^[19]. AWP uses ERC-20 Transfer logs. Both streams are wallet-filtered to the cohort under study: a log is retained if the owner or spender (approvals) or the sender or recipient (transfers) lies in the seed wallet set. Filtering at query time reduces scanned payload relative to full-chain extracts and focuses the reputation subgraph on economically relevant neighbors of the cohort. Extracts run in six monthly batches to respect per-query byte caps and to allow resumable checkpoints if a month fails.
- **GMX events:** Trading-success and risk proxies are derived from GMX V2 PositionDecrease events emitted by the GMX EventEmitter at

`0xC8ee91A54287DB53897056e12D9819156D3822Fb`

[23]. Traders are identified via the event account field. Realized profit and loss use `basePnlUsd`; position size changes use `sizeDeltaUsd` when size-weighted robustness variants are computed; liquidations are flagged from order-type metadata in the decoded payload. GMX is used because it provides dense, wallet-level realized outcomes on collateralized perpetual trading—a domain-intuition validation frame analogous to Do et al.’s use of inbound transfer activity, but targeted at trading success rather than passive inflows [16],[53]. Lending-protocol default labels are excluded because under-collateralized lending volume on Arbitrum within scope is insufficient for stable wallet-level labels.

- **Matched cohort definition:** The primary alignment sample is the intersection of (i) wallets with at least three GMX `PositionDecrease` closes in the observation window and (ii) wallets with non-zero connectivity in the reputation subgraph induced by approval or transfer edges. The minimum-close threshold $m = 3$ excludes one-off traders whose success rates are dominated by sampling noise, while remaining inclusive of active retail participants. The connectivity requirement excludes wallets that appear in GMX events but have no approval or transfer edges touching the reputation extract, for whom social PageRank scores would be uniformly zero and uninformative. After BigQuery extraction and preprocessing, this matched wallet cohort has size $n = 5,521$. Both `EndorseRank` and `AWP` are scored on the same seed set so that differences in alignment and runtime are attributable to ranking logic and graph structure, not to inconsistent sampling.
- **What is not in scope:** The study does not ingest mempool data, off-chain order books, centralized-exchange account maps, or cross-chain bridges. It does not attempt entity resolution across multiple addresses controlled by one actor. It does not use proprietary GMX indexer APIs as the primary source of truth; the public log table is authoritative, and decoding is performed in the open pipeline.

3.3 Three-phase collection strategy

Data acquisition and evaluation proceed in three phases that separate expensive remote scans from local, repeatable analysis. The separation is intentional: once parquet artifacts exist, alignment tables, benchmarks, and robustness sweeps can be regenerated without further cloud cost, which is essential for third-party replication and reanalysis.

- **Phase 1 (GMX validation stream):** Extract and decode all GMX `PositionDecrease` logs from the `EventEmitter` over the full observation window. The extract is not wallet-filtered a priori because eligibility is defined from this stream: every decrease in the window is a candidate label. After decoding, wallets with fewer than three closes form the GMX-eligible set. This stream supplies trading-success, inverse-risk, and liquidation-family proxies. The matched wallet cohort used for primary alignment ($n = 5,521$) is the GMX-eligible set intersected with non-zero reputation connectivity after Phase 2.
- **Phase 2 (reputation streams):** For the GMX-eligible set, extract wallet-filtered ERC-20 `Approval` and `Transfer` events in monthly batches. The matched wallet cohort is then the subset with non-zero approval or transfer connectivity. Each month is an independent query with its own dry-run estimate and checkpoint, so a failure in one month does not invalidate prior months. Preprocess approvals to latest allowances per owner–spender–token: given multiple approvals for the same triple, only the last timestamp in the window survives, and a final zero allowance removes the edge. Preprocess transfers to decimal-adjusted event tables with timestamps, preserving sender, recipient, value, token, and block time. These tables are the sole inputs to `EndorseRank` and AWP graph construction for the primary comparison.
- **Phase 3 (local evaluation):** From cached parquet, compute reputation ranks, five primary proxy families, Spearman and Kendall alignment, computational benchmarks, and scaling/robustness extensions. Supplemental active wallets for the expanded pool are drawn from reputation-subgraph addresses (and optional supplemental parquet) without requiring a full-chain re-scan for every robustness experiment. No additional BigQuery scans are

required once the Phase 1 and Phase 2 extracts and any supplemental active-wallet parquet are available.

- **Cost guardrails:** Every remote query begins with a mandatory dry-run estimate of bytes processed before extraction. Each query is aborted if the estimate exceeds a 150 GiB per-query cap.¹ Operators must pass an explicit confirmation flag before paid scans proceed. Extraction metadata—wallet counts, row counts, bytes processed, and timestamps—is logged in the study’s acquisition record (Appendix 8.1). These controls implement responsible use of public cloud resources and make the acquisition cost of the study auditable.

3.4 Pipeline architecture

Figure 3.1 summarizes the end-to-end flow from public logs to dissertation tables. Remote stages write immutable parquet; local stages are idempotent given those inputs and a single versioned configuration file.

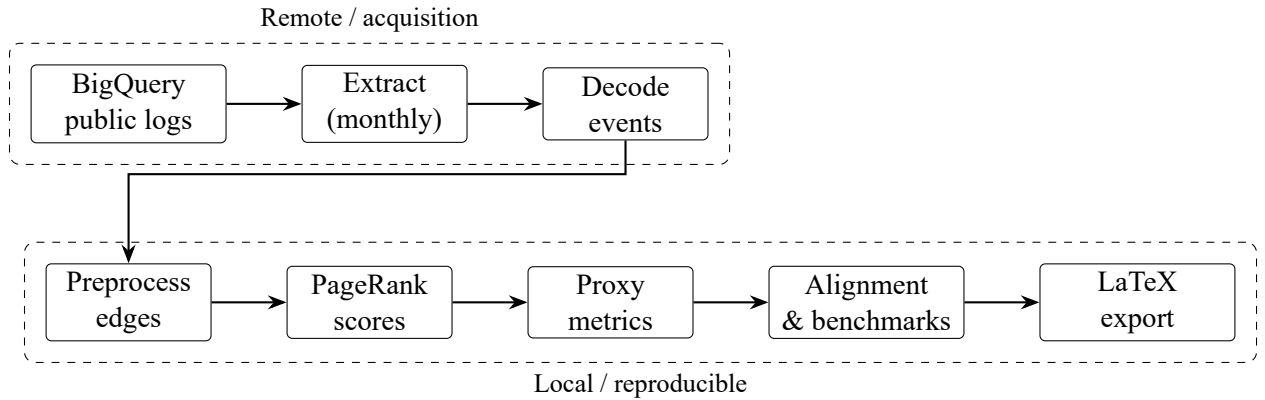


Figure 3.1: Evaluation pipeline: public Arbitrum logs are extracted and decoded, reputation edges and GMX outcomes are preprocessed, EndorseRank and AWP ranks are computed, proxies and alignment statistics are evaluated, and LaTeX result fragments are exported.

Operationally, Phase 1 scripts write GMX raw logs and decoded events; Phase 2 scripts write monthly approval/transfer shards and consolidated latest-allowance and transfer-event tables;

¹The per-query byte cap is an operational policy of this study’s acquisition pipeline, not a property of BigQuery itself.

Phase 3 scripts read those tables, build graphs, run PageRank, compute proxies, and export the result tables reported in Chapter 4. The same pipeline supports offline verification on synthetic fixtures without cloud credentials.

3.5 Event decoding

Blockchain logs store events as topics and a data payload rather than as named fields. Decoding maps each log row to a typed record used by graph construction and proxy computation. Incorrect decoding would silently corrupt both reputation graphs and validation labels; this section therefore records the exact topic hashes and field mappings used in the pipeline.

- **Log anatomy:** An Ethereum-compatible log contains an emitting contract address, an ordered list of topics, and an ABI-encoded data blob. Topic 0 is the keccak-256 hash of the event signature string, which uniquely identifies the event type among well-known standards. Indexed parameters occupy subsequent topics as 32-byte words (addresses are left-padded); non-indexed parameters occupy the data field in ABI order^[18]. BigQuery exposes these fields as columns; the pipeline applies topic filters in SQL and performs ABI decoding in Python for fields that require numeric interpretation.
- **ERC-20 Approval:** Topic 0 is

`0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925,`

the hash of `Approval(address,address,uint256)`. Topics 1 and 2 hold the owner and spender addresses (left-padded); the allowance amount is ABI-decoded from the data field and scaled by the token's decimals. Unlimited approvals (`type(uint256).max`) are retained as large finite weights after decimal adjustment; they contribute strong endorsement edges, which is intentional: an unlimited approval is a strong on-chain authorization. EIP-2612 `permit` does not introduce a separate reputation event: successful permits emit the same `Approval` log, so `EndorseRank` observes the resulting allowance snapshot regardless of whether authorization was set by an on-chain `approve` call or an off-chain signature^[19].

- **ERC-20 Transfer:** Topic 0 is

`0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef,`

the hash of `Transfer(address,address,uint256)`. Sender and recipient are indexed; value is decimal-adjusted. Mint and burn patterns that use the zero address are retained if they touch cohort wallets, but they are rare relative to wallet-to-wallet transfers in the filtered extract. Self-transfers are retained in the event table but excluded from Sybil-stability inbound-counterparty calculations (Section 3.9), because a wallet transferring to itself does not evidence an external counterparty relationship.

- **GMX PositionDecrease:** GMX V2 emits structured events through `EventEmitter` rather than a minimal ERC-20-style signature [23]. Extraction filters on the emitter address and the verified `EventLog1` topic 0, then decodes key-value fields from the event payload to recover `account`, `basePnlUsd`, `sizeDeltaUsd`, and liquidation indicators. Only decreases (closes, partial closes, and liquidations) enter the validation frame; increases open or enlarge positions and do not realize PnL for the success proxies used here. Liquidation flags distinguish forced closes from voluntary ones, which matters for both inverse-risk proxies and GF-PR sink weights.
- **Address normalization and units:** All addresses are lower-cased before joins so that checksummed and non-checksummed hex forms collide correctly. Token amounts are stored as floating-point decimal-adjusted values for weighting; ranking uses relative magnitudes within a cohort rather than absolute USD conversion for allowance and transfer edges. GMX PnL fields are already denominated in USD units in the protocol’s event schema and are used directly for trading-success and risk proxies.

3.6 Graph construction

`EndorseRank` and `AWP` share the `PageRank` solver in Equation (3.1) but differ in edge semantics. Both methods induce a directed, weighted graph on wallet nodes, restrict attention to the subgraph

touching the seed wallet set (edges with at least one endpoint in the cohort, including one-hop neighbors), and emit a score for every cohort wallet (missing nodes receive score 0 before ranking). Subgraph restriction is important for scalability and for interpretability: the score of a cohort wallet may receive mass from non-cohort neighbors that endorse or transfer to it, but the evaluation reports scores only on $\mathcal{W}$.

The conceptual distinction is endorsement versus flow. An allowance is an explicit, revocable authorization to spend; a transfer is a completed movement of value that may reflect payment, routing, market-making, or wash activity without a trust commitment^{[16],[18]}. EndorseRank therefore treats the latest allowance snapshot as a citation-like endorsement graph in the sense of Page et al.^[49]. AWP treats historical transfers as time-decayed citations, down-weighting older flows so that recent economic activity dominates.

3.6.1 EndorseRank allowance graph

Let $\mathcal{A}$ be the set of latest allowance records after preprocessing: for each token and owner–spender pair, only the chronologically last `Approval` in the observation window is retained (zero allowances remove the edge). Across tokens, weights are summed:

$$w_A(u, v) = \sum_{\text{token}} a^*(u, v, \text{token}), \quad (3.3)$$

where a^* is the latest decimal-adjusted allowance. The directed edge $u \rightarrow v$ encodes owner u endorsing spender v . PageRank damping is $d = 0.85$. No temporal decay is applied: allowance override semantics replace time-based weighting, because a newer approval supersedes an older one for the same pair^[18].

Algorithm 1 formalizes construction and scoring.

Algorithm 1 EndorseRank graph construction and scoring

Require: Latest allowances $\mathcal{A}$, seed wallets $\mathcal{W}$, damping d , tolerance τ_{tol} , max iterations I_{max}

Ensure: Scores $\mathbf{s}_{\text{ER}}$ on $\mathcal{W}$

```
1:  $E \leftarrow \emptyset$ 
2: for each  $(u, v)$  in unique owner–spender pairs in  $\mathcal{A}$  do
3:    $w \leftarrow \sum_{\text{token}} a^*(u, v, \text{token})$ 
4:   if  $w > 0$  then
5:     add edge  $(u \rightarrow v, w)$  to  $E$ 
6:   end if
7: end for
8:  $E \leftarrow$  edges in  $E$  incident to at least one wallet in  $\mathcal{W}$ 
9:  $\mathbf{s} \leftarrow \text{WeightedPageRank}(E, d, \tau_{\text{tol}}, I_{\text{max}})$ 
10: return  $\mathbf{s}_{\text{ER}}[w] \leftarrow \mathbf{s}[w]$  for  $w \in \mathcal{W}$  (else 0)
```

3.6.2 AWP transfer graph

AWP follows Do et al.^[16]: each transfer of value v at age Δt days before observation end T_{obs} contributes a decayed weight

$$\sigma(\Delta t) = \frac{1}{1 + \exp(k(\Delta t - t_0))}, \quad k = 0.01, \quad t_0 = 180 \text{ days}. \quad (3.4)$$

Edge weights aggregate over transfers:

$$w_T(u, v) = \sum_{e: u \rightarrow v} v_e \sigma(\Delta t_e). \quad (3.5)$$

Damping is again $d = 0.85$. Figure 3.2 plots $\sigma(\Delta t)$ over the relevant age range: recent transfers retain nearly full weight, while transfers older than the midpoint t_0 are smoothly down-weighted.

Algorithm 2 formalizes construction and scoring.

Because G_{approve} records durable authorizations and G_{transfer} records historical value flows, the two graphs differ in density and temporal sensitivity. Approvals are relatively rare, high-intent events; transfers are frequent and often mechanical. On the matched cohort, the approval subgraph is substantially sparser than the transfer subgraph (Chapter 4), which is the structural basis for the

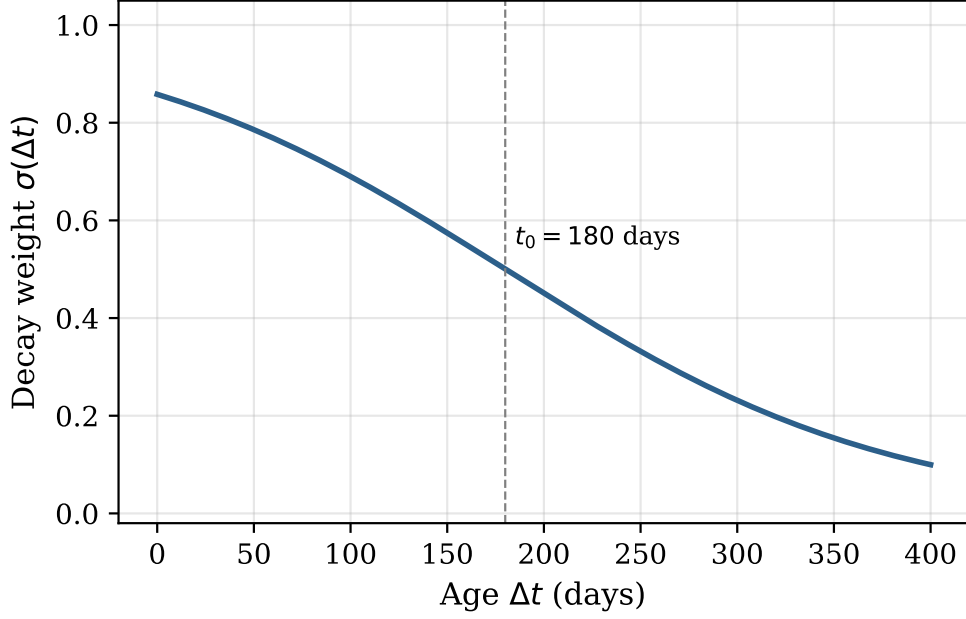


Figure 3.2: Logistic time-decay $\sigma(\Delta t)$ used by AWP ($k = 0.01$, $t_0 = 180$ days). Horizontal axis is event age in days before 2026-05-31 UTC; vertical axis is the multiplicative weight applied to transfer value.

Algorithm 2 AWP graph construction and scoring

Require: Transfers $\mathcal{T}$, seed wallets $\mathcal{W}$, observation end T_{obs} , parameters $k, t_0, d, \tau_{\text{tol}}, I_{\text{max}}$

Ensure: Scores $\mathbf{s}_{\text{AWP}}$ on $\mathcal{W}$

- 1: $E \leftarrow \emptyset$
 - 2: **for** each transfer $e = (u, v, v_e, t_e)$ in $\mathcal{T}$ **do**
 - 3: $\Delta t \leftarrow (T_{\text{obs}} - t_e)$ in days
 - 4: $w_e \leftarrow v_e \cdot \sigma(\Delta t)$ with σ as in Equation (3.4)
 - 5: accumulate w_e on edge $(u \rightarrow v)$
 - 6: **end for**
 - 7: remove non-positive edges; retain edges incident to $\mathcal{W}$
 - 8: $\mathbf{s} \leftarrow \text{WeightedPageRank}(E, d, \tau_{\text{tol}}, I_{\text{max}})$
 - 9: **return** $\mathbf{s}_{\text{AWP}}[w] \leftarrow \mathbf{s}[w]$ for $w \in \mathcal{W}$ (else 0)
-

computational-efficiency hypothesis (H3): per-iteration PageRank cost scales with $|E|$, so fewer edges imply faster solves and lower memory for the adjacency structures in Equation (3.2).

- **Implementation notes shared by both methods:** Edge lists are stored as triples (`from_node`, `to_node`, `weight`) in memory. Duplicate edges are aggregated by summing weights before the solve. Non-positive weights are dropped. The solver builds a sparse compressed-row transition matrix, iterates Equation (3.1), and returns a dictionary from node identifier to score. Cohort scores are aligned to $\mathcal{W}$ with missing entries set to zero, then converted to dense ranks for correlation.

3.7 Supplementary GF-PR graph (extension)

GF-PR is an outcome-native reference built only from GMX `PositionDecrease` events. It is evaluated solely for supplementary question SRQ1 (construct-overlap diagnostic) and is not a primary peer to `EndorseRank` or `AWP`. The method exists to answer a methodological question: if a ranking is allowed to read the same PnL stream that defines trading-success proxies, how high can alignment rise? That ceiling clarifies how much of a social method’s trading-success τ might be attributed to genuine social signal versus residual correlation with outcomes.

The graph is a directed star with two synthetic nodes, POOL and SINK. For each close of wallet w with realized PnL x (field `basePnlUsd`):

- if the close is a liquidation, add weight $|x| + \varepsilon_{\text{liq}}$ on $w \rightarrow \text{SINK}$ with $\varepsilon_{\text{liq}} = 1.0$;
- else if $x > 0$, add weight x on $\text{POOL} \rightarrow w$;
- else if $x < 0$, add weight $|x| + \varepsilon_{\text{loss}}$ on $w \rightarrow \text{SINK}$ with $\varepsilon_{\text{loss}} = 0.1$.

The additives $\varepsilon_{\text{loss}}$ and ε_{liq} break ties when PnL is exactly zero after absolute value and ensure that liquidations contribute strictly more sink mass than ordinary losses of equal magnitude. There are no wallet-to-wallet edges: GF-PR does not encode social endorsement or transfer structure. PageRank with $d = 0.85$ propagates mass from POOL toward profitable wallets and from loss-making wallets toward SINK. High alignment with trading-success proxies is expected by construction and

must be interpreted as construct overlap, not as superior social reputation ^[14].

3.8 Five-proxy evaluation design

Primary evaluation compares EndorseRank and AWP only. For each method M (EndorseRank or AWP) and each proxy P , the pipeline computes scores $\mathbf{s}_M$ and proxy values $\mathbf{p}_P$ on the matched cohort, then reports Spearman’s ρ and Kendall’s τ between the induced rankings (Section 3.10). All proxies are oriented so that higher values imply higher expected reputation before correlation, following the domain-alignment approach of Do et al.^[16]. Orientation is part of construct validity: without it, a method that correctly ranks safer traders could appear negatively aligned with a loss-magnitude proxy.

Five families structure the primary comparison. They were chosen to span (i) baselines from prior work, (ii) within-construct checks for each social graph, and (iii) outcome-oriented trading and risk measures that are not direct inputs to EndorseRank or AWP:

1. **Transfer** (Do et al. baseline): inbound transfer activity.
2. **Allowance**: inbound approval activity (wallet as spender).
3. **Inverse risk** (GMX PnL): loss avoidance and non-loss close behavior.
4. **Sybil-adjusted stability**: counterparty diversity and activity tenure.
5. **Trading success**: GMX close outcomes and realized gains.

Transfer and allowance families are *graph-adjacent*: they are computed from the same event classes that build G_{transfer} and G_{approve} , but they are simple local statistics (degree and value sums) rather than global PageRank scores. Strong alignment of AWP with transfer proxies, or of EndorseRank with allowance proxies, is therefore evidence of within-construct coherence, not of independent predictive power. Inverse-risk, Sybil-stability, and trading-success families are the more demanding tests for social reputation as a signal of trading behavior and account quality.

Family-level mean τ summarizes construct alignment for the EndorseRank–AWP comparison; the evaluation pipeline records which social method aligns more strongly per family (Appendix 8.1).

A sixth liquidation family is computed in the codebase for completeness and is mentioned where relevant, but main-text claims use the five families above.

Supplementary evaluation includes GF-PR only for SRQ1, to diagnose construct overlap on trading-outcome proxies. Lending-protocol default labels remain out of scope. Packin and Lev-Aretz^[48] caution that decentralized credit scores can become opaque; the proxy design counters opacity by publishing explicit formulas (Section 3.9) and by separating social methods from outcome-native references.

3.9 Operational definitions of proxies

Let $\mathcal{W}$ be the matched cohort. For GMX-derived proxies, only wallets with at least $m = 3$ closes in the decoded event table are eligible; ineligible wallets receive zeros after left-join to $\mathcal{W}$. Notation below matches the proxy-computation module of the open evaluation pipeline.

3.9.1 Transfer family

Let $\mathcal{T}_{\text{in}}(w)$ be inbound transfers to wallet w (any sender). Define

$$\text{in_degree}(w) = |\{ u : \exists \text{ transfer } u \rightarrow w \}|, \quad (3.6)$$

$$\text{in_value}(w) = \sum_{e: \cdot \rightarrow w} v_e. \quad (3.7)$$

These are the domain-intuition baselines of Do et al.^[16]: economically important addresses tend to receive transfers from more counterparties and in larger aggregate value. Unlike PageRank, in_degree and in_value are strictly local: they ignore the reputation of senders. Comparing AWP (global, time-decayed) to these local baselines tests whether propagation and recency weighting add ordering information beyond raw inflows.

3.9.2 Allowance family

Let $\mathcal{A}_{\text{in}}(w)$ be latest allowances in which w is the spender (owner $\rightarrow w$). Define

$$\text{in_approve_degree}(w) = |\{u : a^*(u, w, \cdot) > 0\}|, \quad (3.8)$$

$$\text{in_approve_value}(w) = \sum_{u, \text{token}} a^*(u, w, \text{token}). \quad (3.9)$$

These proxies measure endorsement received, the construct EndorseRank is designed to rank. Strong alignment of EndorseRank with this family is expected and is interpreted as within-construct validity rather than independent trading skill. As with transfer baselines, the allowance proxies are local: they count and sum inbound approvals without propagating through multi-hop endorsement chains. EndorseRank’s incremental value, if any, appears when it aligns with non-allowance families while remaining coherent with allowance baselines.

3.9.3 Inverse-risk family

Let $\mathcal{C}(w)$ be the set of non-missing GMX closes for wallet w , with PnL x_c and liquidation flag $\ell_c \in \{0, 1\}$. A loss close is a non-liquidation close with $x_c < 0$. Define

$$\text{loss_avoidance}(w) = - \sum_{c \in \mathcal{C}(w)} \min(x_c, 0), \quad (3.10)$$

$$\text{non_loss_close_rate}(w) = 1 - \frac{|\{c \in \mathcal{C}(w) : \ell_c = 0 \wedge x_c < 0\}|}{|\mathcal{C}(w)|}, \quad (3.11)$$

$$\text{worst_close_pnl_score}(w) = - \min_{c \in \mathcal{C}(w)} x_c. \quad (3.12)$$

Equation (3.10) sums absolute loss magnitudes over all negative-PnL closes, including liquidations (higher is better after the sign flip). Equation (3.11) is the fraction of closes that are not ordinary (non-liquidation) losses; liquidations therefore raise non_loss_close_rate and are not treated as ordinary losses. Equation (3.12) rewards wallets whose worst close is less severe (again after sign flip). Liquidation closes are included in the worst-close minimum but are excluded from the ordinary-

loss count in Equation (3.11), so that forced liquidations are not double-counted as ordinary losses. Inverse-risk proxies are not inputs to EndorseRank or AWP; alignment here is evidence that social structure co-varies with risk-avoiding trading behavior.

3.9.4 Sybil-adjusted stability family

Self-transfers ($u = w$) are excluded from inbound counters. Let $N_{\text{tx}}(w)$ be the number of inbound transfers to w from other addresses and $N_{\text{uniq}}(w)$ the number of distinct inbound counterparties. Let $t_{\min}(w)$ and $t_{\max}(w)$ be the first and last timestamps at which w appears as sender or recipient in the transfer table, and let $M(w)$ be the number of distinct calendar months of activity (inbound months and full-span months, taking the maximum). Define

$$\text{inbound_counterparty_ratio}(w) = \frac{N_{\text{uniq}}(w)}{\max(N_{\text{tx}}(w), 1)}, \quad (3.13)$$

$$\text{transfer_tenure_days}(w) = \frac{t_{\max}(w) - t_{\min}(w)}{86400} \quad (\text{seconds to days}), \quad (3.14)$$

$$\text{active_months}(w) = M(w). \quad (3.15)$$

High counterparty ratio penalizes wallets that receive many transfers from few addresses (a simple Sybil/wash pattern). Tenure and active months capture persistence of on-chain activity within the observation window. These measures are imperfect Sybil defenses—coordinated adversaries can still diversify counterparties—but they operationalize the intuition that durable, multi-counterparty activity is more reputation-relevant than bursty self-dealing ^[48].

3.9.5 Trading-success family

A successful close is a non-liquidation close with $x_c > 0$. Define

$$\text{close_success_count}(w) = |\{c \in \mathcal{C}(w) : \ell_c = 0 \wedge x_c > 0\}|, \quad (3.16)$$

$$\text{realized_gain_proxy}(w) = \sum_{c \in \mathcal{C}(w)} \max(x_c, 0), \quad (3.17)$$

$$\text{close_success_rate}(w) = \frac{\text{close_success_count}(w)}{|\mathcal{C}(w)|}. \quad (3.18)$$

The primary definition of `realized_gain_proxy` is the raw sum of positive PnL. A size-weighted variant multiplies each positive PnL by $\max(|s_c|, 1)$, where s_c is the close's `sizeDeltaUsd`, and is used only in robustness checks (Section 3.12). These proxies operationalize trading success on GMX V2 perpetual markets ^{[23],[53]}. They are outcome measures, not social-graph features: neither `EndorseRank` nor `AWP` reads `basePnlUsd` when building edges. Consequently, non-trivial alignment with trading success is the strongest form of domain evidence available in this design short of external credit labels. Count, sum, and rate capture complementary aspects of success—frequency, magnitude, and consistency—so family-level mean τ is not dominated by a single operationalization.

3.9.6 Liquidation family (ancillary)

For completeness, the pipeline also computes

$$\text{liquidation_free_rate}(w) = 1 - \frac{\sum_c \ell_c}{|\mathcal{C}(w)|}, \quad (3.19)$$

$$\text{zero_liquidation_flag}(w) = \mathbf{1} \left[\sum_c \ell_c = 0 \right], \quad (3.20)$$

$$\text{liquidation_free_closes}(w) = \sum_c (1 - \ell_c). \quad (3.21)$$

These metrics support sensitivity analyses and GF-PR diagnostics but are not part of the five-family primary claims.

3.10 Statistical tests

Reputation scores and validation proxies are continuous rank variables, not binary default labels. Following Do et al.^[16] and the construct-validity tradition of Cronbach and Meehl^[14], alignment is measured with rank correlations that are invariant to monotonic transforms of either variable.

- **Spearman’s ρ :** Let R_i and S_i be the dense ranks of score and proxy for wallet $i = 1, \dots, n$. Spearman’s coefficient is the Pearson correlation of these ranks:

$$\rho = \frac{\sum_{i=1}^n (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum_{i=1}^n (R_i - \bar{R})^2 \sum_{i=1}^n (S_i - \bar{S})^2}}. \quad (3.22)$$

Spearman captures monotonic agreement: if higher reputation consistently co-occurs with higher proxy values, ρ is positive even when the relationship is nonlinear in the raw scores.

- **Kendall’s τ :** Kendall’s coefficient counts concordant and discordant pairs. For all pairs $i < j$,

$$\tau = \frac{2(C - D)}{n(n - 1)}, \quad (3.23)$$

where C is the number of pairs ordered the same way by both rankings and D is the number ordered differently (ties handled by the implementation’s standard tie correction). Kendall’s τ is more interpretable as a pairwise ordering probability and is less sensitive to large rank displacements than Spearman in some heavy-tailed settings.

- **Why both:** Spearman summarizes global monotonic association; Kendall emphasizes pairwise order fidelity. Reporting both follows Do et al.^[16] and allows readers to verify that family-level conclusions are not an artifact of a single rank statistic. In blockchain reputation settings, score distributions are often heavy-tailed: a few hubs receive large mass while many wallets share near-zero scores. Rank correlations remain well-defined in that regime, whereas

Pearson correlation on raw scores would be dominated by hubs.

- **Family aggregation:** Family-level summaries use mean τ across proxies within a family. For example, the transfer-family mean is $\frac{1}{2}(\tau_{\text{in_degree}} + \tau_{\text{in_value}})$. Method comparisons on the same cohort and proxies are relative, not absolute claims about universal reputation. Chapter 2 notes interpretive bands used in related PageRank studies ($|\tau| < 0.10$ negligible; 0.10–0.35 weak-to-moderate; ≥ 0.35 substantively meaningful); these bands guide discussion but do not replace the primary EndorseRank-versus-AWP contrast on identical inputs.
- **Orientation and missingness:** Proxies are pre-oriented so that larger values indicate better reputation. Wallets absent from a proxy source receive zeros after alignment to $\mathcal{W}$, which is conservative for sparse activity and is applied identically to both methods. No wallet is dropped from one method but retained in the other: the correlation sample is always the full matched cohort for primary tables.
- **What is not tested:** The study does not claim causal identification (reputation does not “cause” trading success in an interventional sense), does not fit predictive classifiers with train/test splits, and does not report p -values as the primary decision criterion. The scientific target is comparative construct alignment and computational cost under a fixed, transparent protocol.

3.11 Computational benchmark (Claim C1)

Claim C1 concerns computational efficiency of EndorseRank versus AWP on identical inputs. The benchmark protocol isolates PageRank solve cost from one-time graph construction, because construction cost is dominated by I/O and preprocessing that would be amortized in a deployed scorer that updates incrementally, whereas the iterative solve is the recurring computational core of both methods.

- **Timed quantity:** For each method, edge lists are built once per cohort outside the timed loop. The timed section runs weighted PageRank only, with damping $d = 0.85$, tolerance $\tau_{\text{tol}} =$

10^{-8} , and at most $I_{\max} = 300$ iterations. These numerical settings match the configuration used for alignment scores, so runtime and alignment refer to the same mathematical object.

- **Repetitions and metrics:** Each configuration is repeated for five independent runs to average out operating-system noise. Per run, the pipeline records wall-clock solve time, peak memory, and iterations to convergence. Reported statistics are the mean runtime, runtime standard deviation, mean iterations, maximum peak memory across runs, and edge/node counts of the pre-built graph. Edge counts are reported because they are the leading structural explanation for runtime differences under $O(|E|)$ per-iteration complexity ^[49].
- **Cohorts:** The primary benchmark uses the matched wallet cohort ($n = 5,521$). Scaling benchmarks use deterministic subsamples of the expanded wallet pool (Section 3.12). GF-PR runtime may be reported parenthetically in Chapter 4 as supplementary context only and is not part of Claim C1.
- **Fairness of comparison:** Both methods use the same solver, tolerance, iteration cap, damping (unless a robustness sweep varies d), and seed wallet set. Differences therefore reflect graph density and weight structure—allowance snapshots versus time-decayed transfers—rather than implementation asymmetry. Hardware and software versions are recorded in the evaluation manifest so that absolute seconds can be interpreted relative to the experimental machine; speedup ratios (AWP time divided by EndorseRank time) are the preferred comparative summary.

3.12 Scaling and robustness extensions

Beyond the matched cohort, the study evaluates runtime scaling on a larger address set and stress-tests alignment under parameter and sample perturbations. These extensions support H3 (efficiency at larger n) and the stability of family-level conclusions. They are not alternative primary samples: unless otherwise noted, alignment claims in Chapter 4 refer to the matched cohort, and scaling claims refer to the expanded pool.

3.12.1 Expanded wallet pool and runtime scaling

The expanded wallet pool $\mathcal{W}_{\text{pool}}$ unions the matched GMX cohort with supplemental active addresses derived from the reputation subgraph (and, when present, supplemental active-wallet parquet), yielding $N = |\mathcal{W}_{\text{pool}}| = 31,612$ addresses in this study. Supplemental addresses increase graph size for timing experiments without requiring every address to carry GMX labels. Runtime benchmarks at intermediate sizes use deterministic SHA256 subsampling with seed string `benchmark-tier2-v1`: wallets are ordered by

SHA256(seed || address) with seed `benchmark-tier2-v1`

and the first n addresses form the subsample. The seed string is a fixed configuration constant. Reported stages include $n = 10,000$ and the full pool $n = N = 31,612$, always on the same approval and transfer parquet. In-cohort scaling stages with $n \leq 5,521$ are reported in Appendix 8.1. Deterministic hashing avoids the non-reproducibility of pseudorandom draws across library versions.

3.12.2 Damping sensitivity

PageRank damping is swept over $d \in \{0.75, 0.85, 0.95\}$ on the matched cohort, holding graphs and proxies fixed. The default $d = 0.85$ matches the classical web PageRank setting^[49] and Do et al.^[16]. Lower d increases teleportation and flattens scores; higher d emphasizes multi-hop structure and can slow convergence. The sweep tests whether allowance- and transfer-family alignment are artifacts of a single teleportation weight.

3.12.3 Top-token subgraph

A restricted subgraph retains only the top-20 ERC-20 tokens by combined transfer and allowance volume within the observation window. EndorseRank and AWP are recomputed on this subgraph to assess sensitivity to long-tail tokens and possible spam assets. If family-level conclusions reverse when low-volume tokens are removed, the primary result would be fragile to token composition; stability supports interpreting the full-token graphs as representative of major economic activity.

3.12.4 Sample-size sensitivity

Alignment statistics are recomputed on staged subsamples drawn from the expanded wallet pool using the same deterministic seed. Subsamples include supplemental addresses without GMX labels (zeros on outcome proxies), so reported sample-size τ is interpreted primarily for transfer and allowance families on $\mathcal{W}_{\text{pool}}$; primary claims remain on the matched wallet cohort ($n = 5,521$). Because rank correlations have sampling variability, modest movement in τ across sizes is expected; the robustness question is whether EndorseRank and AWP preserve their relative ordering on key graph-adjacent families.

3.12.5 GMX proxy definition variants

For trading-success robustness, `realized_gain_proxy` is computed under both the primary raw-sum definition (Equation (3.17)) and a size-weighted definition that multiplies positive PnL by $\max(|s_c|, 1)$ for close size s_c (`sizeDeltaUsd`). Size-weighting up-weights large positions and can change the identity of top traders. Primary tables use the raw-sum definition; size-weighted results appear only in robustness tables.

3.13 Reproducibility

All empirical tables in Chapter 4 are produced by an open evaluation pipeline. After Phase 1–2 parquet artifacts are present, a single evaluation driver recomputes proxies, alignment statistics, and benchmarks from those inputs. Command-line flags select cached BigQuery-derived inputs versus synthetic fixtures, enable damping, top-token, sample-size, and GMX-proxy variant sweeps, and export the result tables reported in Chapter 4. Observation window, damping, decay parameters, topic hashes, and benchmark seeds are collected in a single versioned configuration file. Fixture mode supports offline verification without cloud credentials.

Reproducible elements include: deterministic wallet subsampling via SHA256 seeds; fixed PageRank tolerance and iteration caps; explicit proxy formulas in Section 3.9; and logged extraction

manifests. Peer reviewers can regenerate alignment and benchmark tables from the same parquet without re-querying BigQuery.

3.14 Sampling considerations and missingness

The matched cohort is a purposeful sample, not a random sample of all Arbitrum addresses. Eligibility requires GMX activity and reputation-subgraph connectivity, so results speak to *active perpetual traders with observable approval or transfer structure*. Idle wallets, pure holders, and addresses that interact only through non-ERC-20 mechanisms are out of scope by design.

Missingness arises in several forms. Tokens with unknown decimals may be dropped or defaulted during preprocessing; such decisions are logged in the pipeline. Approvals that never emit standard ERC-20 logs (non-standard tokens) are invisible to EndorseRank. Transfers that occur off-chain or through custodial systems never enter AWP. These missingness patterns bias both methods toward on-chain, standard-token activity—acceptable for an on-chain reputation study, but relevant when generalizing to the full user population of a protocol.

Supplemental addresses in the expanded wallet pool ($N = 31,612$) increase coverage for runtime tests but do not automatically carry GMX labels. Alignment claims therefore remain anchored to the matched cohort ($n = 5,521$), while scaling claims use the pool.

3.15 Implementation notes

PageRank is implemented with sparse adjacency lists and power iteration in Python, with optional NumPy vectorization for score updates. Graphs are built once per configuration; timed benchmarks exclude I/O and preprocessing. Randomness enters only through optional fixture generation; real-data evaluation is deterministic given parquet inputs and configuration seeds. Floating-point differences across machines are expected to be negligible relative to the reported τ precision (three decimal places).

Figures are generated from the same evaluation artifacts as the tables, so reported numerics and

plots remain consistent.

3.16 Ethics

This study uses exclusively publicly available blockchain data and does not intervene with human participants. Ethics clearance and data-handling procedures were approved by the institution prior to empirical data collection; the exemption or clearance confirmation is retained as part of the research record.

- **Privacy and anonymity:** Arbitrum One addresses are pseudonymous identifiers, not inherently linked to real-world identities ^[48]. Pseudonymity is not anonymity: determined adversaries can sometimes link addresses to persons through exchange deposits, public profiles, or clustering heuristics. This study deliberately refrains from such linkage. Collection is restricted to on-chain events—Approval, Transfer, and GMX PositionDecrease—with no scraping or integration of off-chain personal data such as IP addresses, exchange KYC records, social-media handles, or deanonymization graphs. No attempt was made to cluster addresses into real-world entities, to publish address-level rankings that could reasonably identify individuals, or to combine wallet histories with external identity providers. Reported results are cohort-level aggregates (correlations, runtimes, family means), not leaderboards of named users. Where examples are discussed in prose, they refer to abstract roles (e.g., “high-allowance spenders”) rather than concrete addresses.
- **Data minimization:** Only fields required for graph construction and proxies are retained in processed tables (addresses, token amounts, timestamps, GMX PnL fields, liquidation flags). Raw logs are stored for auditability of the research pipeline but are not redistributed as a deanonymization aid. Query filters are wallet-scoped where possible to avoid indiscriminate full-chain dumps beyond the study design.
- **Data usage and compliance:** All queries targeted public datasets under `bigquery-public-data.google.blockchain.arbitrum_one_us` in accordance with Google Cloud terms of

service and rate limits. Dry-run byte estimates and per-query caps limit unnecessary scanning. No transactions were broadcast, no smart contracts were called in ways that could disrupt network operations, and no privileged or non-public indexer access was used.

- **Fairness and transparency:** Graph reputation systems can systematically undervalue new entrants, low-volume participants, or wallets that avoid public approvals ^[48]. EndorseRank and AWP were therefore audited for such structural biases: allowance graphs favor addresses that receive spending authority; transfer graphs favor addresses embedded in value-flow networks. Limitations—including cold-start effects, Sybil susceptibility of approval graphs, and construct overlap for GF-PR—are discussed in Chapter 5. All code paths, parameter choices, and evaluation metrics are documented in the open pipeline (Appendix 8.1) to enable peer review and independent replication.
- **Responsible disclosure and impact:** Although EndorseRank may inform risk assessment in DeFi contexts, it is not financial advice, credit advice, or a consumer reporting product. Human oversight remains necessary for lending, trading, and compliance decisions. Potential adversarial manipulation (for example, Sybil attacks on G_{approve} or wash transfers on G_{transfer}) is acknowledged and discussed in Chapter 5. The study reports aggregated cohort-level findings rather than wallet-level rankings intended for targeting individual users. Any future operational deployment would require additional governance, appeal mechanisms, and fairness monitoring beyond the scope of this dissertation.
- **Summary of methodological commitments:** The design fixes the chain, window, cohort rules, graph formulas, proxy definitions, and statistical tests before interpreting results. Primary claims compare only EndorseRank and AWP on the matched wallet cohort; the expanded pool and robustness sweeps test efficiency and stability; GF-PR bounds outcome-native alignment for SRQ1. Chapter 4 reports the empirical outcomes of this protocol.